

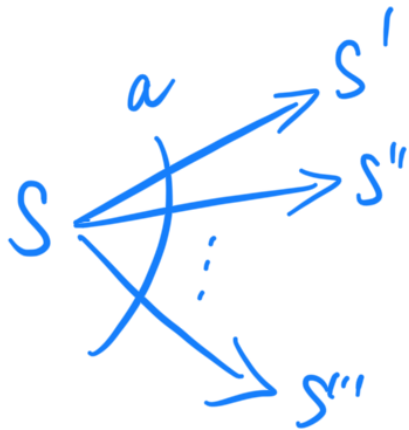
FOND PLANNING

~~CLASSICAL~~

FOND: FULLY OBSERVABLE NON-DETERMINISTIC

all domain properties
are observable
(as in Classical)

actions may
have non-deterministic
effect: multiple successor states



DOMAIN IN CLASSICAL PLANNING

$$D = \langle S, A, \alpha, \underline{\delta} \rangle$$

$$\alpha: S \rightarrow 2^A$$



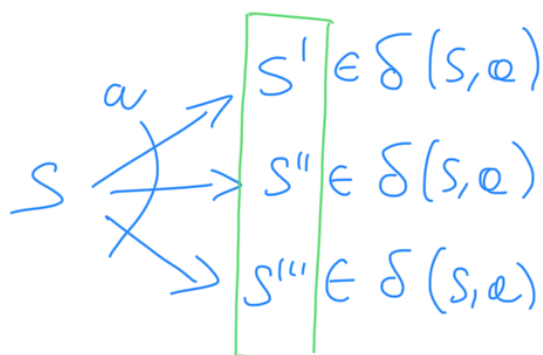
$$\alpha(S) = \{a_1, a_2\}$$

δ : (classical planning) Transition function

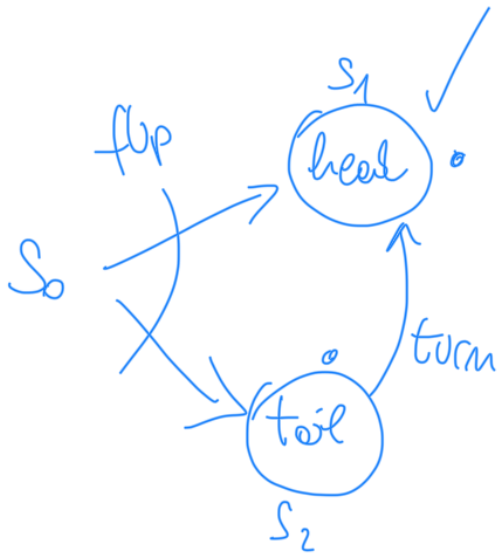
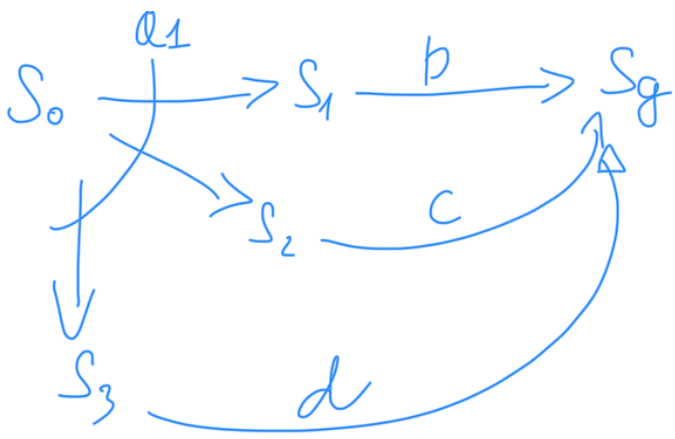
~~$$\delta: S \times A \rightarrow S$$~~

~~$$S \xrightarrow{a} S' = \delta(S, a)$$~~

$$\delta: S \times A \rightarrow 2^S$$



$$\delta(S, a) = \{S', S'', S'''\}$$



$S_0 \rightarrow \text{flip}$
 $S_1 \rightarrow \text{nop}$
 $S_2 \rightarrow \text{turn}$

S	$\pi(S)$
S_0	flip
S_1	nop
S_2	turn

FUNCTION of STATES
 $\neq$
 SEQUENCE of ACTIONS

π : policy $\neq$ plan
 ↑ ↑
 function sequence